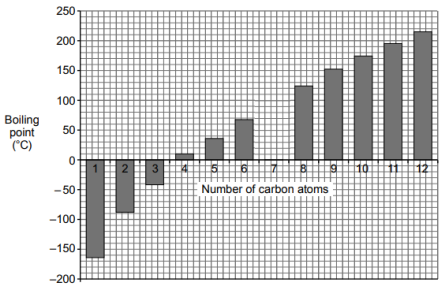
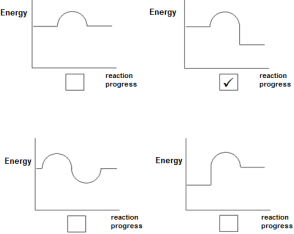


Common questions

| Question |     |     |  | Marking details                                                                                                                                                                                                                                                                                                                                                                  | Marks available |     |     |       |       |      |
|----------|-----|-----|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |     |  |                                                                                                                                                                                                                                                                                                                                                                                  | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 6/1      | (a) |     |  | <p>the <u>larger the molecules</u> / (<u>as the size/fraction</u>) <u>gets bigger</u></p> <ul style="list-style-type: none"><li>the <u>smokier</u> the <u>flame</u> (1)</li><li>the <u>more difficult</u> it becomes to <u>burn</u> (1)</li></ul> <p>accept the converse argument</p> <p>correct identification of both properties, without reference to increasing size (1)</p> |                 |     | 2   | 2     |       |      |
|          | (b) | (i) |  |  <p>all 5 bars correctly plotted with <math>\pm\frac{1}{2}</math> small square tolerance (2)<br/>3 or 4 correct plots (1)</p> <p>accept charts where bars are touching – correct height of each bar to be credited</p>                                                                          |                 | 2   |     | 2     | 2     |      |

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                                    | Marks available |     |     |       |       |      |
|----------|-----|-------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                                    | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |     | (ii)  |  | appropriate <b>straight</b> trend line drawn with ruler, spans across at least 6 bars (1)<br><br>correct boiling point taken from the trend line (1)<br><br>award no credit for a curved trend line (but allow ECF for a correct value read from an incorrect trend line)<br><br>if no trend line drawn, award (1) for a value in the range 85-105 |                 |     | 2   | 2     | 2     |      |
|          | (c) | (i)   |  | sand / foam / CO <sub>2</sub> / fire blanket<br>because it removes oxygen<br><b>both</b> method and explanation needed                                                                                                                                                                                                                             |                 | 1   |     | 1     |       |      |
|          |     | (ii)  |  | 5CO <sub>2</sub> + 6H <sub>2</sub> O<br><br>correct products (1)<br>correctly balanced (1) – only if correct products given                                                                                                                                                                                                                        |                 | 2   |     | 2     | 1     |      |
|          |     | (iii) |  | does not produce carbon dioxide / sulfur dioxide /<br><b>only</b> produces water (1)<br><br>does not contribute to global warming / climate change /<br>acid rain (1)                                                                                                                                                                              | 1               | 1   |     | 2     |       |      |

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks available |     |     |       |       |      |
|----------|-----|-------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (d) | (i)   |  | temperature (of water) <b>before and after</b> burning (1)<br>mass of fuel and burner <b>before and after</b> burning (1)<br><br>initial $\equiv$ before $\equiv$ at start<br>final $\equiv$ after $\equiv$ at end<br><br>temperature rise and change in mass – neutral answers<br><br>award (1) only for reference to measuring both temperature throughout and mass throughout<br><br>award (1) for answers that refer to measuring <b>both</b> the temperature and mass <b>either</b> before <b>or</b> after burning only |                 |     | 2   | 2     |       | 2    |
|          |     | (ii)  |  | distance between burner and flame / material or thickness or size of beaker / same size wick / same beaker<br><br>reference to mass of fuel and mass of water – neutral                                                                                                                                                                                                                                                                                                                                                      | 1               |     |     | 1     |       | 1    |
|          |     | (iii) |  |                                                                                                                                                                                                                                                                                                                                                                                                                                            |                 | 1   |     | 1     | 1     |      |
|          |     |       |  | Question 6/1 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2               | 7   | 6   | 15    | 6     | 3    |